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# An experimental investigation on growth rate of water lettuce (*Pistia stratiotes* L.)

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**Abstract-** *Pistia stratiotes* commonly called water lettuce is free floating aquatic herbs is the member of family areaceae. This is an alien invasive species. This grows very fast and soon covers the entire surface of the water bodies. It is often grown as an ornamental plant in the water bodies or small water tubs owing to its beautiful foliage. But due to its rapid growth it causes it causes nuisance to water bodies as because it is very difficult to control once it cover the entire surface. During the course of study of aquatic and semi aquatic angiosperms of Jharkhand it has been observed that several ponds have been covered with the mat of *Pistia stratiotes*. In order to know its pace of spread, an experiment was conducted at department of botany KKM College, Pakur to observe the multiplication. It was initiated in a water tub. Soon the entire surface of the tub was covered with *Pistia*. In the present communication number in increase of *Pistia* has been recorded on daily basis for a month and same has been discussed.

**Key words: *Pistia stratiotes,* ornamental plant, nuisance, water bodies.**

## INTRODUCTION

*Pistia*, a genus of family araceae is an aquatic floating herb. Commonly called water lettuce or water cabbage. The genus "*Pistia*" is derived from the Greek word "pistos" meaning watery. The specific epithet "*stratiotes*" is derived from a Greek word that means soldier, a possible reference to the species *stratiotes* which contains an aquatic plant species with leaves shaped like swords. It is the single genus of the tribe Pistieae that shows its systematic isolation inside the family. It has only one species i.e., *Pistia stratiotes*, This was first reported from the Nile close to Lake Victoria within Africa. Which further spreaded through naturally or by anthropogenic activities to almost entire tropical and subtropical fresh waterway. Water lettuce is among the world's most dynamic freshwater aquatic plants due to the reason it is considered as highly invasive species. The invasion of this species in the environment leads to serious environmental as well as socio-economic damage to its surrounding the communities. In water with high nutrient load this plants shows luxuriant growth and presents weedy appearance.

Water lettuce is among the world's most dynamic freshwater aquatic plants and is considered as an invasive species. The species can be introduced to new areas by water dispersal, fragmentation, and hitchhiking on marine transportation or fishing equipment.

It reproduces rapidly through vegetative fragmentation with the help of stolon as well as seeds. Under optimal environmental condition in doubles its number in three week. It occurs in the surface of lakes

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and ponds. The heavy growth of this plant hampers the effective uses of water causes clogging of canals and dominates in wetland environments.

**Description of the plant:**

Free floating herbs having 2-10 cm long leaves, surrounded at its base by a membranous sheath, pubescent, obovate-cuneate; apex rounded and truncate. Spathes 10-12 mm long, yellowish unite, obliquely campanulate, contracted near the middle. Carpels obliquely adnate to the spadix; apex free. Ovary one loculed with numerous ovules on a marginal placentae. It flowers and fruit in the month of August - April.

**Systematic position**

| | | |
|---|---|---|
| Kingdom | - | Plantae |
| Subkingdom | - | Viridiplantae |
| Infrakingdom | - | Streptophyta |
| Superdivision | - | Embryophyta |
| Division | - | Tracheophyta |
| Subdivision | - | Spermatophytina |
| Class | - | Magnoliopsida |
| Superorder | - | Lilianae |
| Order | - | Alismatales |
| Family | - | Araceae |
| Genus | - | *Pistia* L. |
| Species | - | *P. stratiotes* |

## MATERIALS & METHOD

One pair of *Pistia stratiotes* was taken and is allowed to grow and multiply in a controlled condition. The increase in number of *Pistia* has been observed on daily basis. The date wise increase in numbers were recorded and photographs of the increased number were taken. On the basis of the same a graph has been plotted.

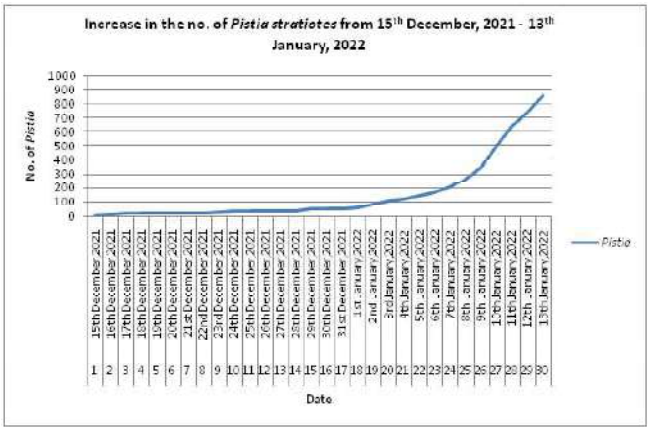
## RESULT & DISCUSSION

One pair of *Pistia* was allowed to grow on a controlled condition in a pot on 15th of December, 2021, very next day i.e. on 16th of December, 2021, the number increased gradually to 11, 18, 18, 18, 18, and then 19, 19, 27, 31, 31, 32, 32, 32 further 46, 46, 49, 54, 80, 100, 120, 140, 165, 200, 255, 300, 500, 640,746 and finally 859 on 13th January, 2022.

This plant being alien invasive soon covers the entire surface, causing disturbance in reaching the sunlight for bottom feeder and alters the grazing the food chain. The huge decomposition of *Pistia* settles in the bottom and reduces the depth.

**Table 1- Datewise multiplication of *Pistia stratiotes***

| Sl. No. | Date | No. of *Pistia* |
|---|---|---|
| 1. | 15th December, 2021 | 2 |
| 2. | 16th December, 2021 | 11 |
| 3. | 17th December, 2021 | 18 |
| 4. | 18th December, 2021 | 18 |
| 5. | 19th December, 2021 | 18 |
| 6. | 20th December, 2021 | 18 |
| 7. | 21st December, 2021 | 19 |
| 8. | 22nd December, 2021 | 19 |
| 9. | 23rd December, 2021 | 27 |
| 10. | 24th December, 2021 | 31 |
| 11. | 25th December, 2021 | 31 |
| 12. | 26th December, 2021 | 32 |
| 13. | 27th December, 2021 | 32 |
| 14. | 28th December, 2021 | 32 |
| 15. | 29th December, 2021 | 46 |
| 16. | 30th December, 2021 | 46 |
| 17. | 31st December, 2021 | 49 |
| 18. | 1st January, 2022 | 54 |
| 19. | 2nd January, 2022 | 80 |
| 20. | 3rd January, 2022 | 100 |
| 21. | 4th January, 2022 | 120 |
| 22. | 5th January, 2022 | 140 |
| 23. | 6th January, 2022 | 165 |
| 24. | 7th January, 2022 | 200 |
| 25. | 8th January, 2022 | 255 |
| 26. | 9th January, 2022 | 350 |
| 27. | 10th January, 2022 | 500 |
| 28. | 11th January, 2022 | 640 |
| 29. | 12th January, 2022 | 746 |
| 30. | 13th January, 2022 | 859 |




**Graph 1- Showing the growth pattern of *Pistia* for one month**




**Fig. 1- Datewise multiplication of *Pistia stratiotes***

As far as the growth and multiplication of the *Pistia* is concerned, so far no study has been conducted, however several work on aquatic flora has been done so far, pioneer among them are as follows:-

The aquatic and semi aquatic flora of district level has been done, studies on aquatic and semi aquatic angiosperms of Lohardaga.[1] Studies on aquatic and semi aquatic angiosperms of Ranchi.[2] Study of aquatic weeds and their control in and around Jamtara district.[3] The floristic and ecological studies of aquatic angiopsperm of Jamtara district and adjoining Barakar basin.[4] Studies on floristic diversity of Jubillee Park of Jamshedpur.[5] The floristic composition and vegetation profile of aquatic angiosperms of Hazaribag and adjacent area.[6] Studies on angiospermic flora of ponds of Ranchi, Jharkhand,[7] Hydrophytes of Ranchi.[8,9] Work on floristic studies of aquatic and semi aquatic angiospermic bio-diversity of Ranchi has also been published.[10] Weeds diversity of Pakur.[11] Floristic studies of aquatic and semi aquatic angiosperms of Ratu Maharaja Pond.[12] Biodiversity of aquatic flora in Raja bandh pond of Jamtara district.[13] Aquatic biodiversity of Victoria Lake of Lohardaga.[14] Studies on the aquatic and semi aquatic angiosperms of Kanke Dam, Ranchi.[15] A survey conducted in the Dumka district reveals the occurrence of 30 species of invasive aquatic and semi aquatic medicinal plant representing 26 genera and 17 families.[16]

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